Unit 9, Lesson 7: Dilations

Benchmark

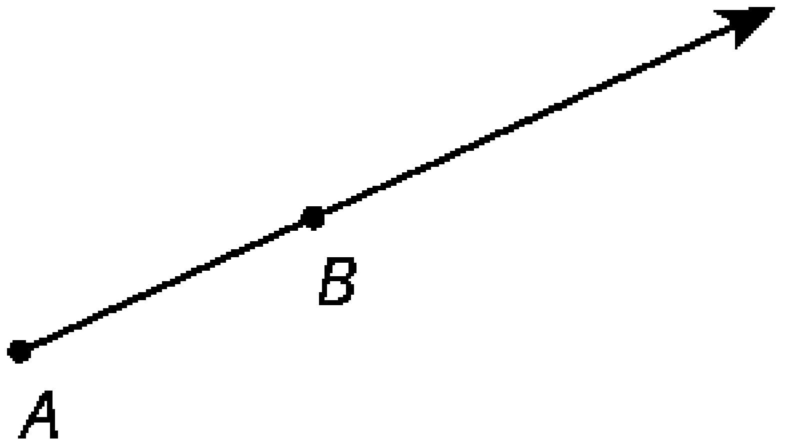
MA.8.GR.2.2 Given a preimage and image generated by a single dilation, identify the scale factor that describes the relationship.

Learning Targets

- ☐ **I can describe the effect on an image given the scale factor.**
- ☐ **I can relate dilations to proportions.**

9.7.1 Warm-Up: Points on a Ray

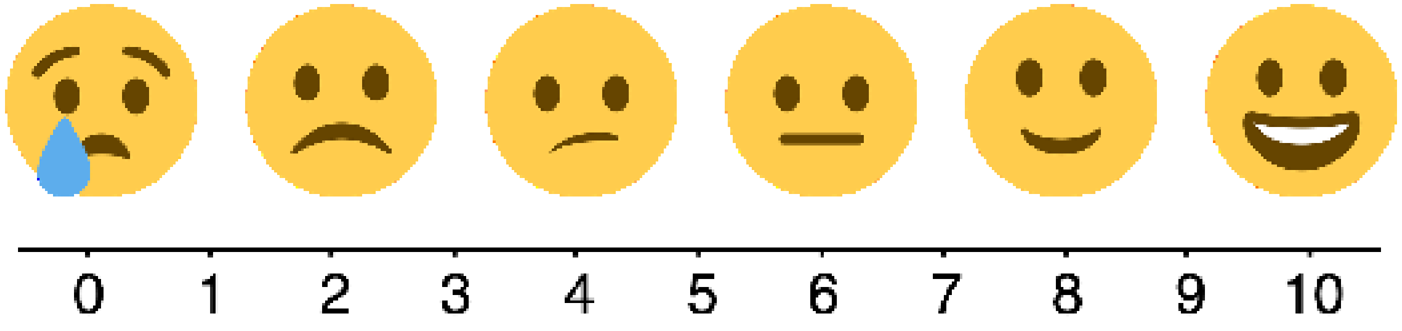
1. A ray is shown with points A and B labeled.



a. Find a point on the ray whose distance from A is twice the distance from B to A. Label the point C.

b. Find a point on the ray whose distance from A is half the distance from B to A. Label the point D.

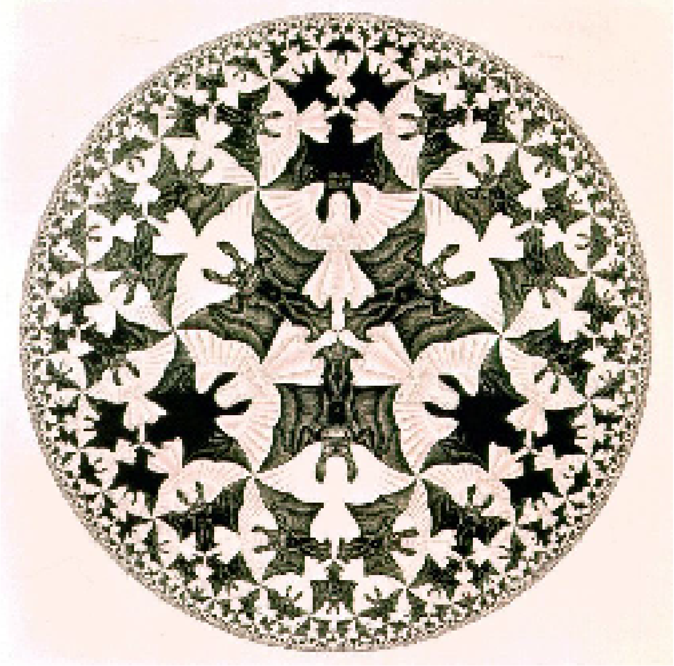
c. Use the emoji scale to rate your confidence in your placement of points C and D on the ray.



9.7.2 Collaboration: Artwork of M.C. Escher

M.C. Escher (1898-1972) was a Dutch artist whose work was inspired by mathematics. Though initially trained as an architect, he became one of the most famous and influential graphic artists in the world.

Below are two examples of Escher's work.

Example A	Example B
	

1. With your partner, discuss how the two example pieces shown are similar and different. Then, record at least two in each column.

Similarities	Differences

2. Consider how the figures used to make the artwork were transformed.

a. Which of the art pieces could be created using only rigid transformations of the shapes used?

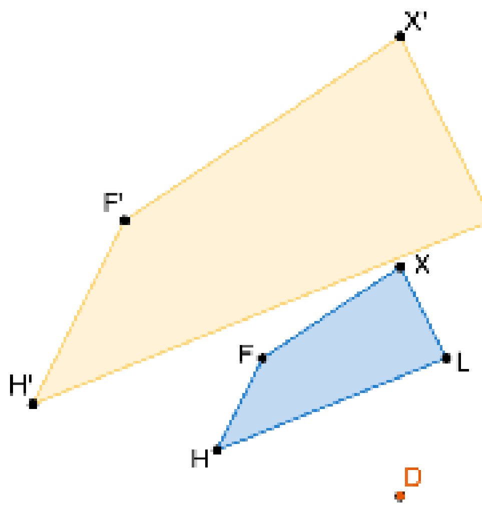
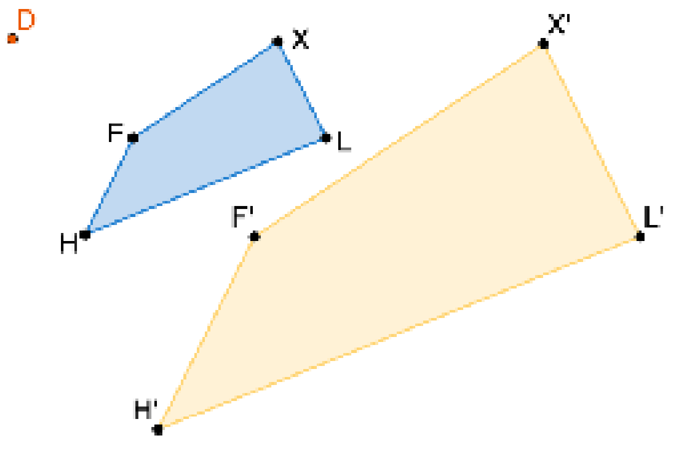
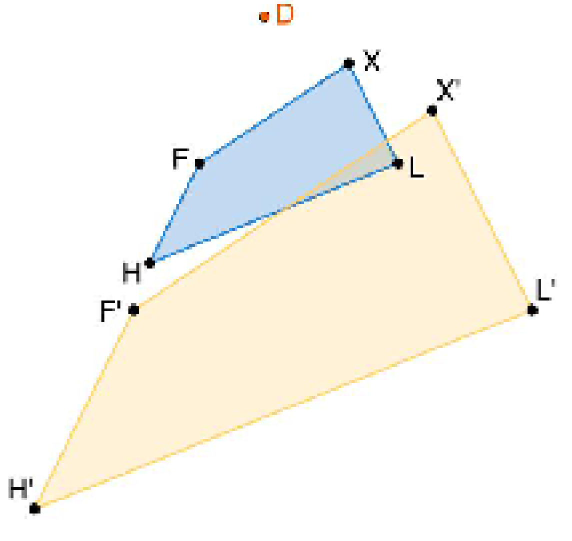
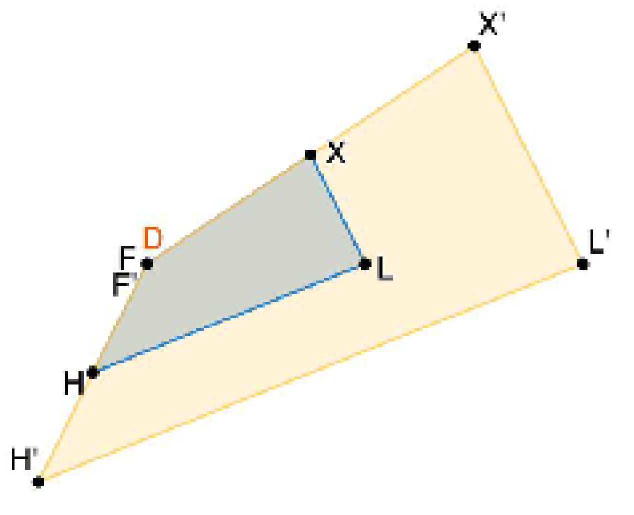
b. Explain how you know the other example could not be created from rigid transformations.

9.7.3 Exploration: Properties of Dilations

Another type of transformation is called a ***dilation***. Dilations are not rigid transformations because the size of the image can change from the preimage.

To dilate a figure, both a center and a scale factor must be given. Work with your partner to complete the following as you explore the properties of dilations.

1. Each of the transformations shown is a dilation of the preimage, quadrilateral $FXLH$, using the same scale factor, but a different center of dilation. The center of dilation is indicated by point D.

Transformation A	Transformation B
	
Transformation C	**Transformation D**
	

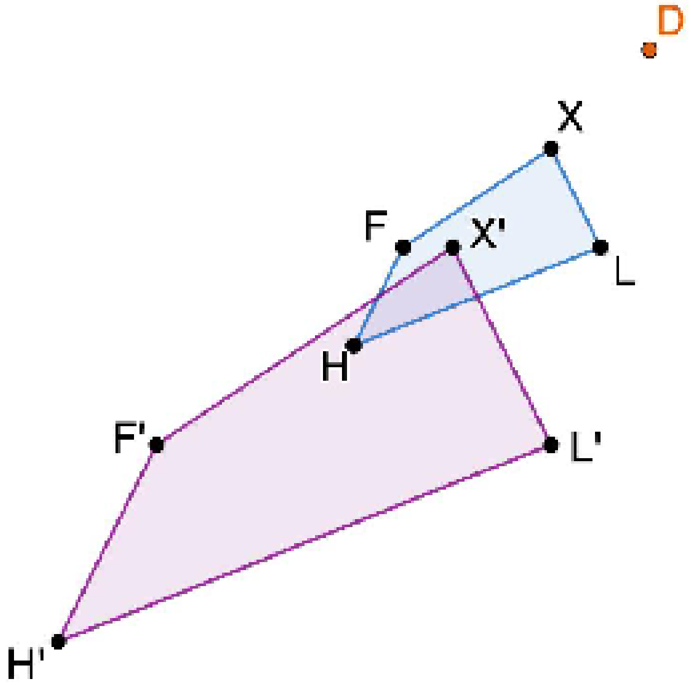
a. Discuss with your partner what you notice about each dilation.

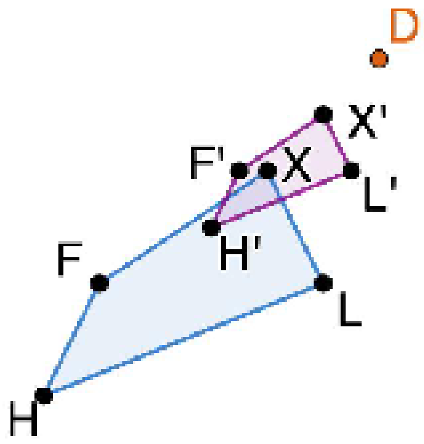
b. Describe what happens to the image, quadrilateral $F'X'L'H'$, as the center of dilation, Point D, changes.

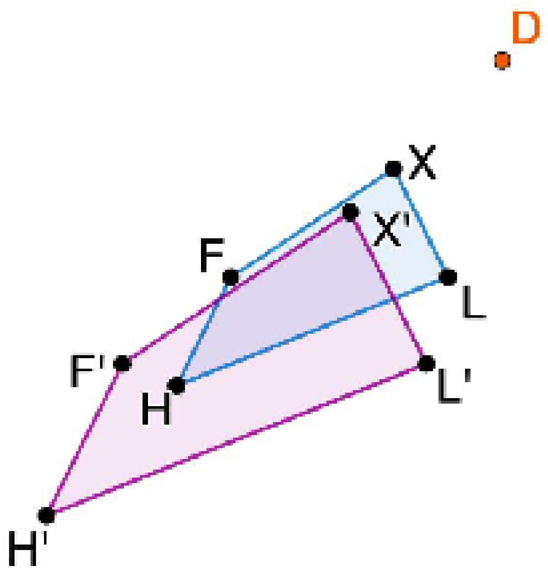
c. In which transformation is the center of dilation a point on the preimage?

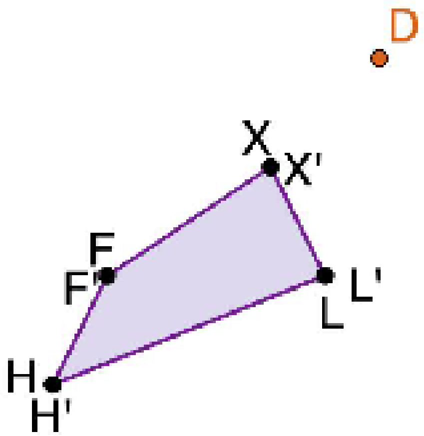
d. Describe how this dilation appears to be different from the others.

2. **Each of the transformations shown is a dilation of the preimage, quadrilateral *FXLH*, using the same center of dilation, but a different scale factor. The center of dilation is indicated by point *D*, and the scale factor is given in the table.**

<table>
<tr><th>Transformation E</th><th>Transformation F</th></tr>
<tr><td>

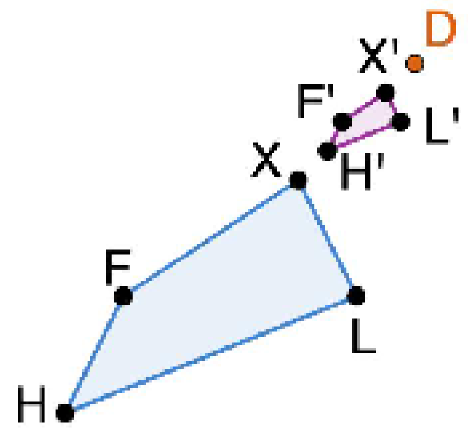
Scale Factor = 2</td><td>

Scale Factor = 0.5</td></tr>
<tr><th>Transformation G</th><th>Transformation H</th></tr>
<tr><td>

Scale Factor = 1.4</td><td>

Scale Factor = 1</td></tr>
</table>

Transformation I



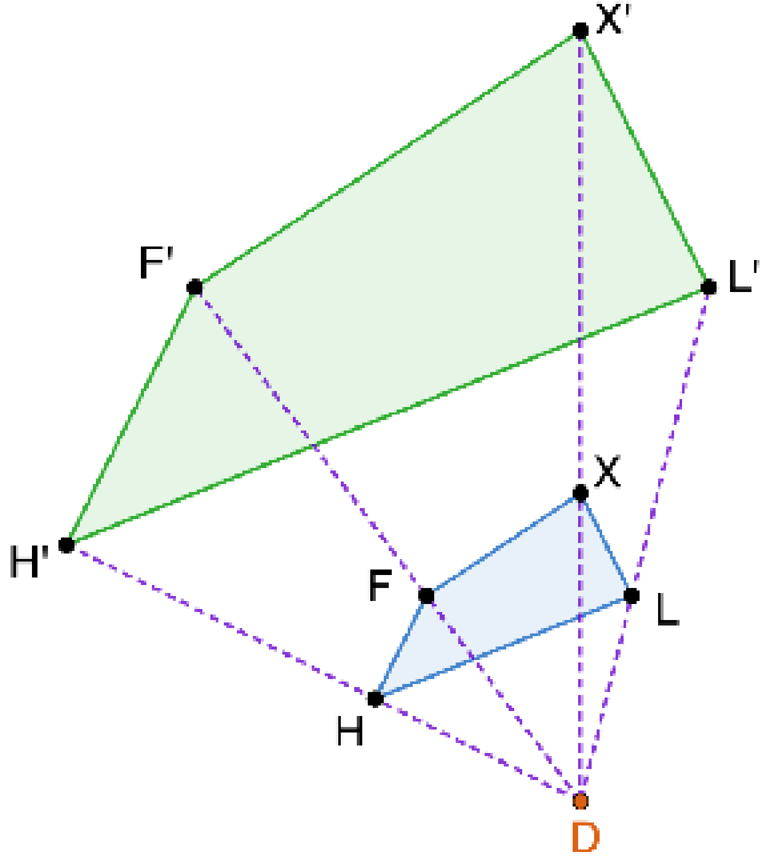
Scale Factor = 0.25

a. Discuss with your partner what happens to the image, quadrilateral *FXLH*, as the scale factor changes.

b. Complete the table.

Value of the Scale Factor	Between 0 and 1	Equal to 1	Greater than 1
How does the image compare to the preimage?			

3. The transformation shows a dilation of quadrilateral $FXLH$ with the center of dilation at point D and a scale factor of 2.5.



a. The distance, in units, between the center of dilation, D, and each of the vertices of the preimage and image are given in the table. Work with your partner to complete the table.

Vertex	Distance from Point D	Ratio $\left(\frac{\text{image}}{\text{preimage}}\right)$ of the Distances
H'	11.25	
H	4.5	
F'	10.25	
F	4.1	
X'	12.5	
X	5	
L'	15	
L	6	

b. **Find the value of each ratio in the table using your calculator. What do you notice?**

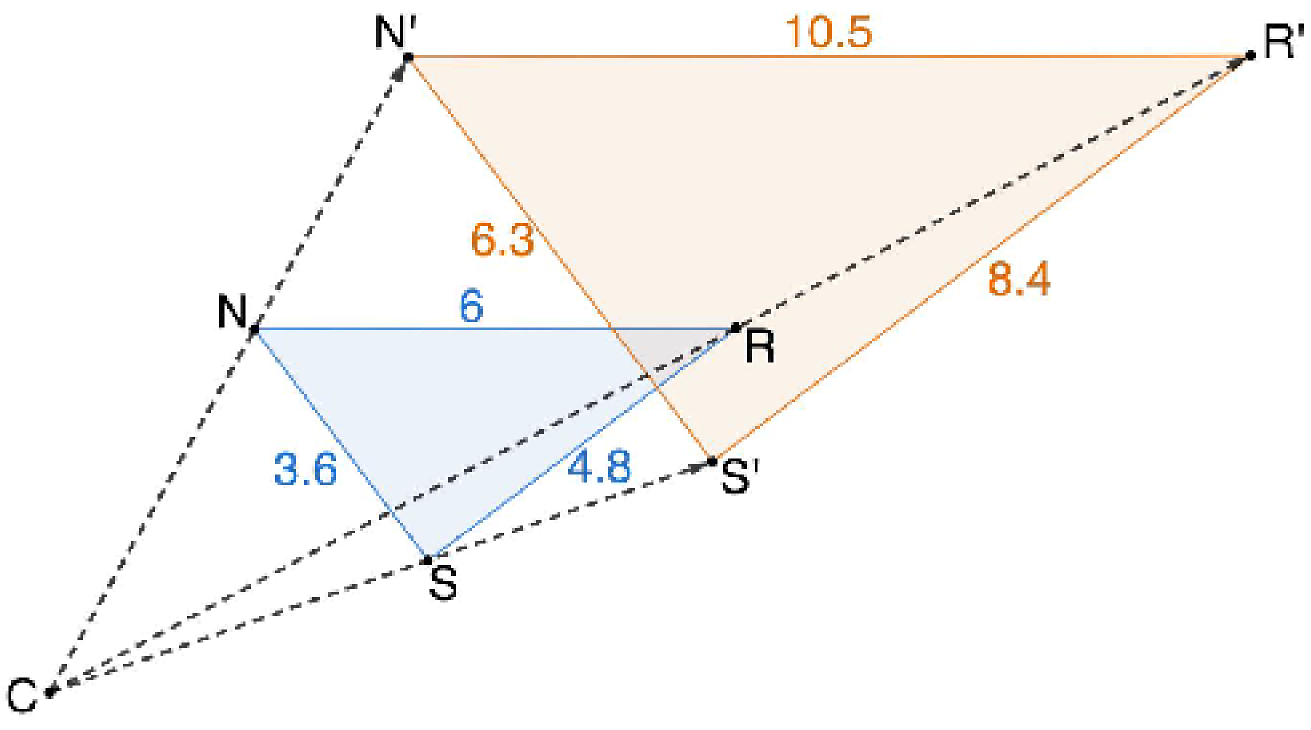
c. **Explain how the ratios of the distances compare to the scale factor of the dilation.**

9.7.4 Bring it Together: Properties of Dilations

Every dilation has a center of dilation and a scale factor.

In a dilation, the distance of the image to the center of dilation is **proportional** to the distance of the preimage to the center of dilation. The constant of proportionality is equal to the **scale factor.**

1. Triangle NRS was dilated using a scale factor of 1.75 centered at point C. The side lengths, in units, of each triangle are shown.



a. Trace ΔNRS on patty paper, labeling each vertex on the patty paper.

b. Overlay the patty paper on $\Delta N'R'S'$ at each vertex to compare the size of each corresponding angle. What do you notice?

c. Complete the table using the side lengths of the image and preimage.

Side	Length	Side	Length	Ratio of the Side Lengths, $\frac{\text{Image}}{\text{Preimage}}$
NR		$N'R'$		
RS		$R'S'$		
SN		$S'N'$		

d. What do you notice about the side lengths of ΔNRS and $\Delta N'R'S'$?

e. Complete the statements using words from the bank.

congruent	proportional	similar

After a dilation, the corresponding side lengths of the preimage and image are ____________________ and the corresponding angles are ____________________. Therefore, the preimage and image are ____________________ figures.

A ***dilation*** is a proportional increase or decrease in size in all directions.

9.7.5 Wrap-Up: Reflection

1. Complete the table.

The terms and information in the Exploration and Bring it Together that I knew were …	Something I wondered during the Exploration or Bring it Together was …	Something I learned during the Exploration or Bring it Together was …